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The Brain's Best Guess

Better a diamond with a flaw than a pebble without.

—Confucius[1]

Le mieux est l'ennemi du bien. [The perfect is the enemy of the good.]

—Attributed to Voltaire[2]

Philosophy is written is this grand book, the universe, which stands continually open to our gaze. But the book cannot be understood unless one first learns to comprehend the language and read the characters in which it is written.

—Galileo Galilei[3]

We have two brains in our skull or at least two virtual divisions. First, there is the "good-enough" brain. This is largely prewired and acts quickly via a minority of highly active and bursting neurons connected by fast-conducting axons and strong synapses into a network. The good-enough brain judges the events in the world in a fast and efficient way but is not particularly precise. The privileged minority of this virtual division is responsible for perhaps half of the spikes in the brain at any given time and share information among themselves and have faster access to the rest of the brain than the remaining majority of neurons. The strongly interactive circuits that form the good-enough brain can generalize across situations but with less

1. https://www.goodreads.com/quotes/63434-better-a-diamond-with-a-flaw-than-a-pebble-without.

2. https://fr.wiktionary.org/wiki/le_mieux_est_l%E2%80%99ennemi_du_bien.

3. https://www.princeton.edu/~hos/h291/assayer.htm.

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than perfect fidelity. In short, the strongest 10% or so of synapses and fast-firing neurons in the brain do the heavy lifting at all times. They do a good-enough job under most conditions, just as a few key players from a large orchestra can produce an enjoyable recital.[4]

The fast execution of plans by the good-enough brain is amply illustrated by Wesley Autrey, the New York City "Subway Hero." He saved the life of a young person who had suffered a seizure and stumbled from the platform, falling onto the tracks in 2007. Without hesitation, Autrey dove onto the tracks and protected the man by throwing himself over his body while the subway cars passed over them. In an interview, Autrey summarized perfectly what I am trying to say about the good-enough brain: "I just saw someone who needed help. I did what I felt was right."[5] All of us do such acts all the time, although in a less heroic form.

However, good enough is far from perfect. We would not want to drive a car with 60–80% accuracy or submit a scientific paper with such precision. To perform better, we also need to deploy the second virtual brain: a large fraction of slow-firing neurons with plastic properties that occupy a large brain volume connected by weaker synapses into a more loosely formed giant network. Their work is absolutely critical for increasing the accuracy of brain performance.

Of course, I am not thinking about two discrete brains in one skull, but instead a continuum of a broad distribution of mixed networks that performs apparently different qualitative computations at the left and right tails. This distribution allows the brain to implement anything from preexisting rigid patterns to highly flexible solutions. Thus, there is no definable boundary between the "fast decision, low precision" and "slow decision, high precision" networks.[6] Brain performance is a tradeoff between speed and accuracy.

After one of my talks on this topic, somebody commented: "The dichotomy you were talking about reminded me of Daniel Kahneman's book *Thinking, Fast and Slow.* This was an embarrassing moment for me. I was familiar with

4. A common observation in physiological experiments is that the activity of just a dozen or so strongly firing neurons is often more informative about an animal's behavior than members of the majority of slow-firing neurons. For example, in brain–machine interface applications, just ten strongly task-related neurons of a hundred recorded neurons can predict 60% accuracy for limb position or gripping force. Adding spikes from the remaining ninety neurons improves the prediction by only a few percent. The "backbone" of task-relevant neurons provides a "best guess" for appropriate action. To increase task performance by another 15% would require a thousand simultaneously recorded neurons, and to achieve 30% improvement would perhaps need ten thousand (Nicolelis and Lebedev, 2009). A key question is what would happen in the network if the committed task-related minority, the backbone, was silenced.

5. https://www.nytimes.com/2007/01/03/nyregion/03life.html.

6. These paragraphs summarize the review by Buzsáki and Mizuseki (2014).

Kahneman's interesting work on cognitive biases with Amos Tversky,[7] but I had never heard of the book. The first thing I did after returning to my hotel room that night was to order the paperback. The Table of Contents did not reveal any obvious relationship to my thesis. On the other hand, the blurb on the back of the book was explicit: the mind has two systems. "System 1 is fast, intuitive, and emotional; System 2 is slower, more deliberative, and more logical." These short sentences induced panic. Is it possible that I had read about this research somewhere, and my poor source memory tricked me into believing that Kahneman's ideas were mine? I left the book on my desk and glanced at it every night but did not have the courage to delve into it. Then, after two weeks or so, I surrendered and read it over a weekend from the beginning to the end. Chapter after chapter, my heart rate went up, as I dreaded finding an outline of how brain mechanisms support the alleged two systems. But there was no mention of the brain.

Kahneman discussed numerous psychological experiments that effectively highlight the differences between his two phenomenologically defined systems and explained eloquently how they could produce different behavioral results even given the same input. After finishing the last page, I made peace with the book and with myself. Kahneman and I arrived at similar conclusions even though we approached the same problem from very different directions. His interest is the mind and cognition, and he took an economics point of view. I started my research program inside the brain and made my way slowly outward to conclude something similar; namely, that the wide distributions of brain dynamics allow it to perform complementary operations that may often appear fundamentally different. These dedicated networks can quickly make good-enough decisions, but precision requires a more protracted process across a large area of the brain. There are no two systems, just one system with two tails. Thus, our views are complementary, and we are looking at the two sides of the same coin.[8] In this last chapter, I will recapitulate the headlines of the previous chapters and attempts to fill the inside-out framework with content.

7. Tversky and Kahneman (1974, 1981). Daniel Kahneman received the Nobel prize in Economic Sciences in 2002. *Thinking, Fast and Slow* (2011) was published three years before our "log-dynamic brain" paper (Buzsáki and Mizuseki, 2014). In my defense, it took us two years of struggle to publish the experimental basis of the log distributions (Mizuseki and Buzsáki, 2013). Kahneman's book won the National Academies Communication Award. It illustrates how our cognitive biases, judgments, and decisions shape everything from planning an experiment to playing the stock market.

8. The two approaches are not just distinct at the level of investigation (psychology vs. neuroscience). Kahneman's approach falls under the umbrella of discrete and nonhierarchical general contrast models (Tversky, 1977), in contrast to the log scale-based continuous distributions I propose here.

PREFORMED BRAIN DYNAMICS

Perhaps no self-respecting neuroscientist subscribes to the *tabula rasa* view, which explicitly states that the mind is a whiteboard on which experiences are written. Yet experiments in most neuroscience laboratories are still being performed according to this outside-in tradition or its modern-day version, *connectionism*.[9] The concept of *tabula rasa* is alive but well hidden.

Perhaps the best illustration of the outside-in perspective is the field of memory research. Nearly all computational memory models tacitly start out as a blank slate onto which new experiences or representations are engraved.[10] Most artificial neural networks suffer from an inconvenient bug known as "catastrophic interference." The failure is called "catastrophic" because new learning can effectively erase all stored memories in connectionist models, which never happens to people (short of severe head trauma). The problem of overwriting remains a significant challenge for the development of computational models.[11] This would be an issue in a hypothetical *tabula rasa* brain as well, which is essentially a connectionist model where the dynamic is built up during the course of learning because adding new information would inevitably destabilize its dynamic state. Furthermore, brain wiring and physiology of members of the same species with different degrees of experience would be dramatically different in *tabula rasa* brains because, in such models, the properties of the network are scaled by the amount of experience. Yet it is not clear how neurons in such a brain would know where to direct attention or what events in the world to process. Equally importantly, there is no easy way to understand how a blank slate, passive observer brain, embedded in an outside-in framework, can become a doer and creator of goals.

9. It has been repeatedly suggested that preconfigured connectivity and an internally regulated dynamic come at a cost of redundancy and constrain the combinatorial possibilities of neurons, mainly based on the observation that improvement of sensory performance is often accompanied by reduced spike correlation among cortical neurons (Cohen and Maunsell, 2009; Mitchell et al., 2009; Gutnisky and Dragoi, 2008). This redundancy is usually referred to as "noise correlation" (Zohary et al., 1994; Averbeck et al., 2006; Cohen and Maunsell, 2009; Renart et al., 2010; Ecker et al., 2010). The term "noise" in the brain, of course, only reflects our ignorance of refined mechanisms.

10. See e.g., Marcus et al. (2014). "Representationism" cannot explain the relationship between "the thing to be represented" and "the thing that represents." Is it one of similarity? Is it metaphoric (symbolic)? Or is it a numerical map, like the representation of pictures in a digital camera? In normal perception, the *tabula rasa* brain is impacted by external objects and perceived by the mind. Illusions are then distorted representations, and hallucinations are unfounded representations (Berrios, 2018).

11. McCloskey and Cohen (1989); McClelland et al. (1995).

Although views about neuronal mechanisms of perception underwent several important transformations over the past two decades, these new developments remained largely within the realm of the outside-in framework. Problems in today's perceptual research have been strongly connected to decision-making and are couched in the language of Bayesian estimation, named after the eighteenth-century thinker Thomas Bayes.[12]

One should distinguish between the "Bayesian method" as a body of mathematics, which can effectively deal with observations and contrast them with outcomes, and the "Bayesian brain model," which is based on a metaphoric interpretation between observations and experimenter-assumed goals of the brain, such as an ideal observer and efficient evaluator of situations. Formally, the Bayesian method is a disciplined statistical method to collect and evaluate data. For example, given the sequential activity of hippocampal neurons during sharp wave ripples during sleep, how well can this observation explain the animal's position in the maze, given the sequence (Chapter 8)? The Bayesian brain model is a representational framework with a strong emphasis on evaluation of perceptual inputs and decision-making with no or very little role attributed to action or internal motor reafferentation to sensory processing.[13] Its predictions rest on an internal or a generative model of how sensory inputs unfold. It posits that the brain—more precisely, our perceptual system—makes assumptions about the objective world. More precisely, it suggests that representation of all possible values of the parameters is computed along with associated probabilities in successive stages, and this process leads to the most efficient prediction. When a decision-maker of the Bayes theorem needs to make a prediction, it factors in the prior outcomes, such as previous successes and failures in similar situations. The Bayes model allows the brain to integrate different sensory cues and modalities gradually and carefully without committing too early to a particular judgment.

The Bayesian model presumes that more complex brains, such as ours, can accurately estimate the properties of the objective world on the basis of sensory information alone. It assumes that sensory information can provide an observer-independent truth, yielding veridical perception.[14] Decisions are

12. Bayes's theorem was generalized and expanded by the French mathematician Pierre-Simon Laplace (in his Analytic theory of probability, 1812) to address problems such as celestial mechanics and medical statistics. The Bayesian inference methods gained enormous popularity in the field of machine learning (e.g., Bishop, 2007; Stone, 2013) and more recently in virtually all fields of neuroscience. Bayesian statistical inference is particularly effective in the dynamic analysis of neuronal sequences (Pfeiffer and Foster, 2013).

13. Knill and Pouget (2004); Doya (2007); Friston (2010); Friston and Kiebel (2009).

14. The "truth" has been a long-standing theme in Western culture and it is believed to be absolute. The truth is either assumed to be objective (out there) or subjective (created by the

made on accumulated evidence that also includes previous experience, and the model tacitly assumes that evolution gradually perfected our perceptual systems. But flies, frogs, mice, and humans may see and evaluate the same environment differently. Thus, postulating that the human's brain view is more veridical than that of other species is hard to justify.

The idea of actively examining the sources and meaning of sensory signals in the Bayesian model resonates with the hypothesis-testing and future-predicting brain view of the inside-out framework. Furthermore, the important role of previous experience and phylogenetic hardwiring ("priors") link aspects of the Bayesian model to the inside-out framework as well. However, in contrast to the Bayesian model, grounding is a key feature in the inside-out framework and is provided by a second opinion of action-based experience. There is no assumption or need that the perceptual system describes the true state of affairs in the outside world. As servants of the action system, the sensors are tuned to effectively guide the animal in its ecological niche by supplying sufficient clues. Sufficiency is not a feature of the objective world but reflects a relationship between the organism's needs and its environment.

Empty Versus Prewritten Book

A view opposite to the *tabula rasa* has long been advanced, perhaps starting with Plato, although he was agnostic about the brain. According to Platonic philosophy, "ideal forms" exist independent of the observer, perhaps not too far from the idea of preexisting brain dynamics. We mortals are chained in a cave, and all we ever perceive of the world and the beings in it are the shadows they cast on the wall of our cave as they pass its entrance. While Platonic and empiricist philosophies are diametrically opposite in many ways, they are virtually identical when it comes to the tacitly assumed source of knowledge: passive observation, either from directly sensing the signals or indirectly observing their shadows. Thus, both views fall under the outside-in framework. To make the preexisting neuronal patterns useful, Plato's cave dwellers must break loose of their chains, leave the cave, explore the world, and get a second opinion. We need to venture out to bring back knowledge. The only way we can learn that sticks that "look bent" in water are not really broken is to touch and move them.

perceiving mind through understanding). However, understanding a situation always depends on our interaction with the situation. Different versions of truth give rise to different accounts of meaning because what one considers as truth always depends on one's conceptual system.

If brain networks and dynamics are preformed, what advantages do they offer over the blank slate model? First and foremost, its preexisting "ideal forms" provide the necessary balance to keep the brain's dynamical landscape stable and robust against other competing needs, such as wide dynamic range, sensitivity, and plasticity (Chapter 12). There is no threat of catastrophic interference because preformed brain networks are not significantly perturbed by new experiences. Indeed, computational models attempting to avoid catastrophic interference include two different synaptic populations. One set can change rapidly but decays to zero rapidly ("fast" weights); the remaining set is hard to change but decays only slowly back to zero ("slow" weights). The weighting used in the learning algorithm is a combination of slow and fast weights, reminiscent of the two ends of the log distribution of synaptic weights in real brains.[15] Second, newly acquired experience is not created in the sense of adding new words to a vocabulary list. Instead, the preformed brain is an already existing dictionary, although its numerous words and sentences are initially meaningless. As a sharp piece of lifeless stone has the potential to become an essential tool for scraping meat or a weapon, neuronal words have the potential to become meaningful to the organism after experience attaches utility to them. To paraphrase Galileo, the book (i.e., the brain) cannot be understood unless one first learns to comprehend its language and read the characters in which it is written. The meaning of the neuronal words in this brain-book is acquired through exploration.

In the preformed brain model, neural words and sentences are neither built from scratch nor fully controlled by perceptual inputs—they preexist. They are combined from preexisting components with a skewed distribution rule. These components, such as neuronal assembles (letters) contained in gamma or ripple cycles (Chapters 4 and 8) are fully active even when the brain disengages from the environment or falls asleep. Evolving cell assemblies, therefore, reflect

15. Hinton and Plaut (1987). In this model, the strong and weak synapses are not distributed, but each connection has two weights: a slowly changing, plastic weight that stores long-term knowledge and a fast-changing, elastic weight that stores temporary knowledge and spontaneously decays toward zero. When old memories are "blurred" by subsequent training on new associations, the memories can be "deblurred" by rehearsing just a few of them. As soon as the fast weights have decayed back to zero, the newly learned associations are restored. Fusi and Abbott (2007) followed a similar rationale, using a model with logarithmic dependence of memory capacity on the number of synapses used to store the memories. A system-level solution to interference is to have a rapid learning structure with a mechanism for rapid acquisition of new information that does not interfere with previously stored information (e.g., the hippocampus). Subsequently, the hippocampus serves as a teacher to the slow-learner neocortex, where details of all existing memories are stored (McClelland et al., 1995; Chapter 8).

the default functional mode of the brain. In fact, it would be hard to imagine nondynamic, stationary circuits in which neurons would be silent or idling.

To summarize this section, I suggest that brains come with a preexisting dynamic even without prior experience, providing a scaffold that allows it to make guesses about the consequences of the actions of the body it controls and to filter which aspects of the world are worth attending. The brain is not a blank tablet to be filled gradually by the truths of the world but an active explorer with a preformed dynamics ready to incorporate events from its points of view. The brain's only job is to assist the survival and prosperity of the body it interacts with, independent of whether, in the process, it learns "objective reality" of the outside world or not.

EXPERIENCE AS A MATCHING PROCESS

For the *tabula rasa* brain, knowledge is synthesized from scratch. For the inside-out model, it is experience that adds meaning to preformed neuronal trajectories and their combinations. We discussed abundant experimental evidence, particularly the skewed distribution of many anatomical and physiological parameters, in previous chapters that favor the latter view. Rich brain dynamics can emerge from the genetically programmed wiring circuits and biophysical properties of neurons during development even without any sensory input or experience. One such fundamental, experience-independent source of dynamics is brain rhythms. The brain's hierarchically related oscillations serve a dual purpose: they maintain stability and robustness on the one hand and offer a needed substrate for syntactical organization of neural words and sentences on the other (Chapter 6). This is the organization I call the *preformed or preconfigured brain*: a preexisting dictionary of nonsense words combined with internally generated syntactical rules. The neuronal syntax with its hierarchically organized rhythms determines the lengths of neuronal messages and shapes their combinations. Thus, brain syntax preexists prior to meaningful content.

Neuronal circuits with recurrent connections, the prototypical organization of the mammalian cortex, cannot be silent for long. Instead, their default dynamic is the maintenance of neuronal sequences through self-organization (Chapter 7). All brain networks are characterized by a rich dynamic of constantly evolving cell assembly sequences that can be combined by the rhythm syntax in a virtually infinite number of ways.[16] This process produces a rich

16. Similar views have been expressed by others. Ken Harris, a previous postdoctoral fellow from my lab, refers to the "realm of possibilities" offered by cell assembly sequences in the

realm of potential neuronal words and sentences, although the words acquire meaning only by action-based experience. Meanings are action-calibrated neuronal trajectories.

Only through actions can neurons relate their responses to sensory inputs to something else supported by the corollary discharge mechanism (Chapters 1, 3, and 5). Mainly through such mechanism can sensory signals acquire meaning, defined as their significance to the organism. Because the brain's action signals are always copied to sensory circuits (Chapter 3), meaningless neuronal words can become meaningful by comparing actions with sensations. The copy of the output provides sensory circuits with a second opinion, a sort of a reality check against what comes into the brain through the sensors. Every self-generated action, even a glance of the eye, can be regarded as the brain's hypothesis testing.

The null hypothesis of brain circuits is that nothing strange happens outside there. However, when the comparator mechanisms detect a difference between action-produced expectation and sensory inputs, the null hypothesis is rejected. This discrepancy triggers further investigation, and those preexisting neuronal patterns that coincide with the attended unexpected event are marked as important. As more knowledge is accumulated, a larger fraction of neuronal trajectories become meaningful to the organism (Figure 13.1). With increasing experience, the comparator mechanisms can boost their ability to notice even more subtle differences. Thus, in contrast to the general wisdom that "everything is novel to a newborn baby," I think the opposite might be the case. Instead of moving from the specific to the general, for a newborn, every human face is just a phylogenetically biased vague face cue. It takes several months of brain training to differentiate among the many possible faces and eventually identify the mother and father as explicit identities and different from everybody else and to distinguish novel from the familiar.

Let me try to explain this idea differently. Whereas it is intuitive for an adult to think that concepts are formed by extracting joint features from individual elements, in reality, we learn categories and gestalt concepts faster than we do specifics. When a child sees a dog and names it as a "dog," it is neither a

cortex (Luczak et al., 2009). George Dragoi, a previous student of mine, calls it "preplay" of future, never before-experienced, sequences (Dragoi and Tonegawa. 2011, 2013a, 2013b; Liu et al., 2018). The similarity between spontaneous and sensory-evoked spatial patterns in the visual cortex of anesthetized monkeys and ferrets has also been interpreted in the framework of preconfigured networks (Kenet et al., 2003; Tsodyks et al., 1999; see also Fiser et al., 2004 and MacLean et al., 2005). My first assignment as a student was to examine the variability of click-evoked responses in the auditory cortex of the cat. The conclusion of the experiment was that the main source of variability of the evoked events is the background spontaneous activity (Karmos et al., 1971).

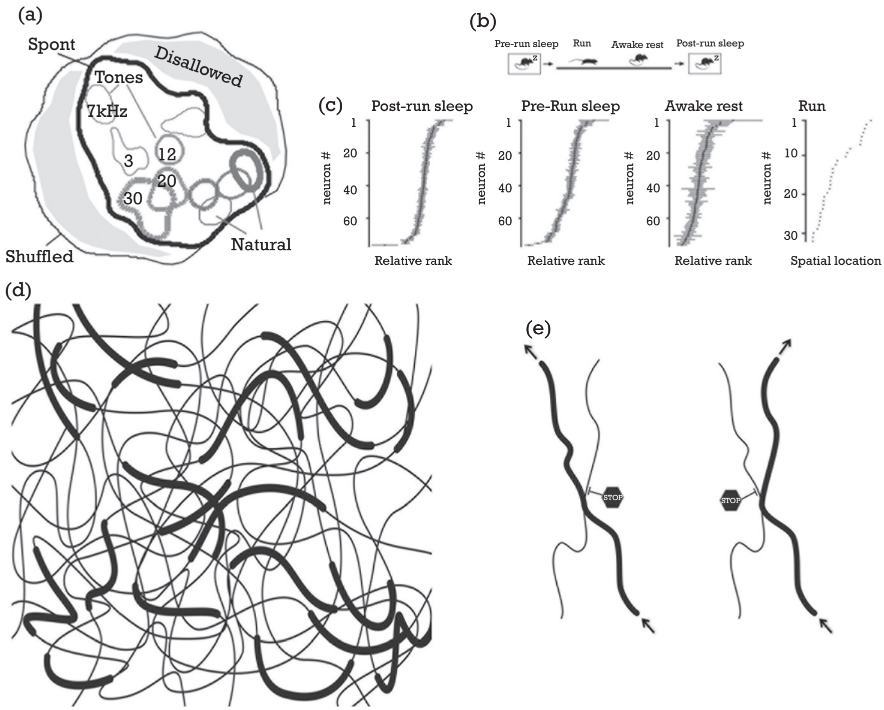

Figure 13.1. Neuronal responses to stimuli and environmental changes are selected from a large reservoir of internally generated neuronal patterns. A: Cartoon illustrating the geometrical interpretation of the relationship between spontaneous and evoked events in the rodent auditory cortex. Of all theoretical scenarios (outer boundary), neuronal connectivity constrains the realm of possible self-organized patterns (area outlined by thick black line). The gray area outside this boundary (disallowed) does not occur spontaneously. Evoked responses to 3–30 kHz tones (outlined by various shapes) and natural sounds (lower, right ovals) are drawn from the preexisting spontaneous patterns. B: Comparison of behavioral states in which similarity of neuronal sequences in the hippocampus were examined: pre-experience and post-experience sleep in the home cage (pre-run and post-run), running on a maze track, and resting. C: Neurons with place fields on the track were ordered according to their place field peaks (*rightmost panel*). Neuronal sequences with the same sequential order as during run were found in abundance not only during resting and post-experience sleep but also during sleep *before* the rat entered the novel track, indicating that the pattern of activity during the very first experience was drawn from a reservoir of self-organized spontaneous patterns. D: Illustration of a small fraction of the large number of possible neuronal trajectories (thin lines). Segments of the trajectories that have already gained meaning to the animal are marked by thick lines (experience-matched trajectories). E: Inhibition (STOP sign) can affect the movement of neuronal traffic and prevent/allow cross-over from one trajectory segment to another.
A: Reproduced with permission from Luczak et al. (2009); B, C: Reproduced with permission from Liu et al. (2018).

specific nor a general category but a mixture. Confronted with a sheep, he may say "dog." A superordinate category, for example "mammal," can be readily matched to a preexisting neuronal trajectory and generalized to many similar appearances. Specifying a subordinate, such as your specific pet dog, needs several extra features which are learned slowly by concatenating preexisting segments of neuronal trajectories (Figure 13.1). The first utterances of a baby are generalized concepts rather than specific references. "Bana" may refer to seeing a banana, requesting a banana, or daddy eating a banana. Linguists call these early generalized utterances "holophrases." The general to specific or good-enough to precise transition is also more economical. It is not realistic to assume that we learn all things in our surrounding world and give them names. Even when we name something (e.g., a house), it is most often vague (e.g., it may or may not include a hut or an igloo). The brain's economic solution seems to be a compromise between precision and the brain's storage and processing capacity.

Even an inexperienced brain has a huge reservoir of unique neuronal trajectories with the potential to acquire real-life significance. But only exploratory, active experience can attach meaning to the largely preconfigured firing patterns of neuronal sequences. The reservoir of neuronal sequences contains a wide spectrum of high-firing, rigid and low-firing, plastic members that are interconnected via preformed rules (Chapter 12). The strongly interconnected preconfigured backbone, with its highly active member neurons, enables the brain to regard no situation as completely unknown. As a result, the brain always takes its "best guess" in any situation and tests its most plausible hypothesis. Each situation—novel or familiar—can be matched with a highest probability neuronal state, a reflection of the brain's best guess. The brain just cannot help it; it always instantly compares relationships rather than identifying explicit features. There is no such a thing as unknown for the brain. Every new mountain, river, or situation has elements of familiarity, reflecting previous experiences in similar situations, which can activate one of the preexisting neuronal trajectories. Thus, familiarity and novelty are not complete strangers. They are related to each other and need each other in order to exist. In case of a discrepancy—when there is a mismatch between an input and the highest probable existing trajectory—a novel combination is constructed from existing trajectories and added to the brain's "knowledge base" (thick lines in Figure 13.1D). For example, after multiple corridors of a maze have already been matched to neuronal trajectories by the process of exploration, a novel combination of a never-taken path between two corridors can be achieved by concatenating the neuronal trajectories of the two corridors (Chapter 8; Figure 13.1E). This process is how I imagine that the initially nonsensical neural words of the brain's vocabulary become meaningful and useful. Meaning is thus neither objective not absolute but relative to the grounding action.

I take the risk of moving my speculation one step further. Once a neuronal pattern acquires meaning through action-mediated grounding experience, its spontaneous or imagination-assisted recurrence can be used as a simulated stimulus. In turn, manipulating such internalized patterns and combining them in various constellations can generate new ideas and abstract constructs, possibly assisted by the mixing-embedding ability of hippocampal sharp wave ripples (Chapter 8) or related mechanisms in the neocortex. This is possible because each pattern in question has already been anchored by active exploration, and each has acquired some utility for the organism. But the resulting abstract constructs also need to be grounded by action to verify the validity of the inference (Chapter 9). Otherwise, the generated ideas may remain in the domain of entertaining fictional thoughts, such as Santa Claus, or pathological ones, such as hallucinations.[17]

CONSTRAINTS HAVE ADVANTAGES

From our discussion of the log rules of connectivity and dynamics (Chapter 12), the constraints on circuit dynamics should not be surprising. Neurons rarely signal anything alone, as any individual neuron functions mainly as part of an ensemble (Chapter 4). Such collective cohesiveness decreases the freedom of assembly members. Self-organized networks create their own constraints and dynamics that guide the evolution of neuronal trajectories rather than being imposed on them by external stimuli (Chapter 5). Neuronal networks with such internally controlled and continually changing dynamics can respond differently to the same sensory stimulus at different times. Emergence of the statistical features of such evolving firing dynamics does not require external control. Sharp wave bursts are present at birth, before any exploratory experience.[18] Their incidence increases modestly postnatally, perhaps due to circuit maturation. Intrinsic processes can also explain why the synaptic strengths and firing rates of neurons are so strongly preserved across behavioral states (Chapter 12). These circuit constraints also influence the order of firing in sequences of member neurons during different brain states and at different temporal scales. For example, during the sharp wave ripples of sleep (Chapter 8),

17. Feinberg (1978).

18. Leinekugel et al. (2002).

the neurons during exploration of a novel environment may become active in a similar order as during sharp wave ripples of the preceding sleep. This arrangement offers the somewhat counterintuitive prediction that sequential dynamics should exist before experience. [19] Indeed, this idea fits comfortably with observations that firing rates, the strength of synaptic connections between neurons, and their order of firing, in general, are impervious to brain state (Chapters 8 and 12). The preconfigured bias of neuronal circuits for certain activity patterns is also demonstrated by the observation that their artificial activation can induce firing sequences that are related to their native sequences.

In further support of preformed brain dynamics, when an adult animal visits a novel environment for the first time, a particular constellation of neurons is selected from the existing repertoire of sequences and maps. The hippocampus and entorhinal cortex always produce a map in any situation, known or unknown. Exploratory locomotion is randomly interrupted by immobility periods associated with patterns of fast neuronal sequences, replaying the past trajectory of the animal or predicting a new one.[20]

Matching preconfigured patterns with experience is not unique to the hippocampus. In the motor cortex, neuronal firing sequences observed after learning a particular movement or intention of movement are remarkably similar to preexisting neuronal patterns produced before learning, thus demonstrating that neuronal populations are constrained to give rise to neuronal sequences from a large reservoir of internally induced patterns. Instead of random and unlimited combinations, a particular preexisting pattern from the large reservoir is assigned to a new movement and perceived pattern, and thereby the neuronal sequence acquires a behavioral meaning.[21] This framework can be contrasted with the

19. In this experiment, CA1 pyramidal neurons were locally activated by optogenetic drive. The induced firing rates and sequences were correlated with those of native rates and sequences, illustrating strong circuit property constraints in any given situation (Stark et al., 2015). This circuit constraint may explain the remarkable observation that place cell sequences in a novel environment can be predicted from the sharp wave ripple-associated "preplay" sequences during sleep *prior* to the experience (Dragoi and Tonegawa, 2011, 2013a, 2013b; Liu et al., 2018).

20. Replay sequences were even observed after the first lap on a novel track (Foster and Wilson, 2006). The center of the current debate is whether the model of the world that informs replay is developed very rapidly, after only one or two experiences (Silva et al., 2015) or whether at least part of the map existed before exploration (Dragoi and Tonegawa, 2011).

21. In these experiments, monkeys were trained to move a cursor on the computer screen by a "brain-machine interface" in which firing patterns in the motor cortex was the input. Thus, the monkey commanded the direction and speed of the cursor simply by intention. After learning, the monkeys produced roughly the same set of activity patterns across all movements as before learning (Golub et al., 2018). These experiments are in support of the model of neuronal group

outside-in *tabula rasa* model, in which neuronal assemblies and sequences are constructed largely from sensory instructions.

Human language metaphorically illustrates the possibilities and constraints of the log-dynamic brain. The more than 6,000 mutually unintelligible languages are spoken by humans exemplify the rich generative abilities of neuronal circuits. A large part of human language is semantic knowledge acquired through externalization and societal interactions (Chapter 9). Any proposition, intention, or thought can be conveyed by any language and readily translated to another one. Yet, irrespective of the meaning of utterances and written words in different languages, all languages have the same basic generative rules (i.e., a preexisting syntax),[22] and guided by the constraints of the hierarchical syntactic organization of brain rhythms with their log rule.

Good-Enough Guesses

The matching process works fastest if it does not have to be exact. Instead of identifying each item separately in all its details, weighting the evidence in a precise manner, and deliberating on the smartest solution, the brain produces a good-enough action by rapidly assessing the relationships among the input variables. The brain senses relationships in the world effectively due to its skewed dynamical organization.

Gestalt psychologists recognized that the mind has an innate ability to perceive patterns based on similarity, proximity, continuity, closure, and connectedness. All these terms describe a relationship.[23] Detailed knowledge of the parts of an image is not necessary to recognize their sum (i.e., the whole). No amount of analysis of the parts in a Jackson Pollock painting would give a clue to why the whole is perceived as it is, as an aesthetic experience derived exclusively from relationships of form and color likely matching the perceiving brain's statistical dynamics.

The illusions discussed in this book also inevitably emerge from the dominance of the good-enough brain, which evaluates the relationship among the constituent parts and ignores often meaningless elements. Only when previously irrelevant details acquire some behavioral significance to the perceiver does the precision brain step in and identify them. Another way to illustrate how

selection or the "neural Darwinism" of Gerry Edelman (1987), a framework closely related to the inside-out approach.

22. Noam Chomsky proposed that a Universal Grammar binds all known languages (1980). The system of brain rhythms may be the foundation of such grammar (Buzsaki 2010; Chapter 6).

23. Eichenbaum and Cohen (2014).

the preconfigured log-dynamic brain organization affects the good enough–precision spectrum is through our sense of quantities.

RELATIONSHIPS AND SENSE OF QUANTITIES

Math enthusiasts often assume that math is as natural as language and therefore should be taught as early as possible in school. Part of the argument is based on behavioral research, which claims that counting does not need a complicated brain because even simple animals have a sense of numbers. Starting with the infamous counting horse Clever Hans,[24] the list of animal species with an alleged ability to do basic math, such as counting, has been increasing, now including chimpanzees, orangutans, macaques, dogs, parrots, New Zealand robins, coots, salamanders, newborn chicks, and even bees. Some behaviorists go so far as stating that counting is innate in many species.[25] This is strange. If the number sense is as natural to us as the sense of color, why did it take a few million years of hominid evolution to conceive the idea of numbers and use them to count anything (Chapter 9)? [26] My math-expert friends would counter that objection by explaining that mathematics is not about memorizing tables but spotting patterns. A person might be terrible at arithmetic, yet her pattern-spotting ability would allow her to judge quantities efficiently. I agree.

The debate about the innate versus acquired form of math may be a reflection of the good-enough versus precision extremes of a log-dynamic brain because two apparently different worlds are lumped into one discipline. As discussed in the previous chapter, ratio judgment is a natural function of neuronal circuits. Every animal, including young children, can effortlessly estimate ratios. Such estimation is not the exact arithmetic of division or multiplication, but, for practical purposes, it is good enough. Estimating the size difference between two antelopes can save a lion quite a bit of extra running and fighting, even if

24. Hans, of course, turned out to be "cheating" by taking cues from his owner or other observers of his "counting" act (Pfungst, 2000).

25. Matsuzawa (1985); Pepperberg (1994); Jordan et al. (2008); Tennesen (2009). See http://www.scientificamerican.com/article/how-animals-have-the-ability-to-count/

26. Stanislas Dehaene, the French mathematician-neuroscientist, believes that mathematics is language-dependent (1997). In this case, there would be no such thing as preverbal or nonverbal counting (Gallistel and Gelman, 1992), and animals other than humans could not have mathematical skills. Others approach the same problem more radically. Animals "never conceive absolute quantities because they lack the faculty of abstraction" (Shepard et al., 1975; Ifrah, 1985).

she does not calculate the exact ratio of 1.6666. The good-enough brain can make fast and efficient judgments about quantities under most conditions, even without any clues about numbers.

Quantity judgment is different from number-based counting. This view explains why monkeys can "count" only up to five, robins look first to the holes with the most mealworms, and newly hatched chicks go looking for the larger number of objects. They do not count but *estimate* ratios. However, the ability to precisely distinguish between 20 and 21 objects is a different thing. The difference between numbers cannot be innately understood; instead, we have to go to school and learn the invented rules and formalized logic.[27]

Linear Versus Log Scales

Giving each number equal importance by spacing numbers equally along a line is a convention that people agreed upon. This aspect of math is not natural to the brain. Numbers, such as "threeness" and "fourness," are explicit abstract symbols, a human invention that must be learned through hard education. This convention is a contrast to the logarithmic number line in the brain, in which bigger quantities gets progressively more compressed.[28]

In support of the log versus linear division, brain imaging studies show that the estimation aspect of number sense is present from an early age, as reflected by activation of the parietal cortical area. Recall from Chapter 10 that this region overlaps or is remarkably close to areas that represent spatial and geometrical dimensions such as magnitude, location, gaze direction as well as duration. Neurological patients with brain damage in this area show not only spatial neglect but also an idiosyncratic mental number line. For example, they might choose five as the midpoint of a numerical interval between two and six, perhaps biased by the brain's natural tendency to estimate logarithmic values.[29] In contrast to good-enough estimation, problems

27. Geary (1994); Gardner (1999); Devlin (2000).

28. Humans without formal math education, such the Munducuru, an Amazonian indigene group, map symbolic and nonsymbolic numbers onto a logarithmic, instead of linear, scale (Dehaene et a., 2008). Their performance is somewhat similar to that of Western kindergarteners, who also devote more space to small numbers, imposing a compressed logarithmic mapping. For instance, they might place the number 10 near the middle of the 0 to100 segment (Siegler and Opfer, 2003). My guess is that children's difficulties with education-based arithmetic may result from their persistent reliance on preexisting logarithmic representations of numerical magnitudes.

29. Brannon and Terrace (1998); Zorzi et al. (2002); Sawamura et al. (2002).

requiring exact calculation activate the frontal cortical regions, including language areas.[30] Thus, "natural numbers" are not so natural to the brain. They and their combinatorial rules have to be acquired through social interaction, similar to language.

In line with the imaging studies, electrophysiological studies in rhesus monkeys also demonstrate that firing of single neurons in the parietal and prefrontal cortex obeys a log rule of quantity representation. The experimenters 'asked' the monkeys to judge whether consecutive images containing one to five dots showed the same or different numbers of dots. There were two surprising findings. First, many neurons responded most strongly to pictures containing the same number of dots (e.g., 2 or 4), irrespective of their sizes and positions. Second, when neuronal firing responses were plotted as the function of the log number of dots, they conformed to a Gaussian curve with a fixed variance, obeying the Weber-Fechner law. Thus, this experiment finds, again, that neuronal firing corresponds to the perceived quantities on a logarithmic rather than by a linear scale.

MEANING IS ACQUIRED THROUGH EXPLORATION

One can argue that acquisition of new knowledge, such as a map, is a result of generalization from previous experience with similar situations. What about the brain of a newborn animal?

Rats, even more than humans, have limited sensory abilities at birth. Rudimentary sniffing and whisking can be observed at postnatal day 6, and by day 10 they reach adult levels. During the first week of life, pups can only crawl. Ambulation on four limbs emerges only by the end of the second week of life, after which the pups begin to leave the nest. The length of such exploratory trips increases abruptly at the beginning of the third week, and the pups become independent of their mother by the end of the third week.

The head direction system is already stable early in life. The first exploratory trip beyond the confines of the nest takes place between postnatal day 15 to 17. At this stage of development, hippocampal pyramidal neurons already display adult-like place fields on the very first trip and ordered sequences, while emergence of stable entorhinal grid cells is delayed until weaning. Thus, the hippocampus possesses many possible neuronal trajectories before exploration has begun. Because of these preformed trajectories, new maps can be retrieved

30. Dehaene et al. (1999).

instantly when pups enter into novel environments.[31] Hippocampal neurons do not fire because the cues at particular parts of the environment make them fire but because the internally evolving neuronal trajectory reaches the point that it is their turn to become active. Think of a dictionary of artificial words. We can map English words to this artificial language by randomly opening a page in the dictionary and blindly pointing to a word.[32] The knowledge of the English words can then ground the meaning of the artificial words, similar to the grounding ability of action-based experience. Of course, the brain is not a simple dictionary that can be opened randomly at any page. It has built-in constraints, as we discussed in Chapter 12. Thus, there seems to be a preorganized priority to which experience can be linked.

The first spatial map may be only a crude, yet good-enough sketch (a "protomap")—the brain's best guess about the new situation. With further exploration, plastic neurons from the large reservoir of the slow firing majority can be recruited to the protomap (Chapter 8). Under this admittedly speculative framework, the physiological correlates of learning from experience involve rapid matching to the familiar aspects of the environment based on preexisting dynamics of the fast-firing minority of neurons, followed by gradual refinement of firing patterns in a subgroup of the slow-firing reserve population to reflect the novel aspects of the situation. For example, the cycles of a sharp wave ripple event contain a sequential series of cell assemblies, each of which can be conceptualized as a letter of the hypothetical neuronal vocabulary so that multiple cycles form a neuronal word. In clusters of sharp wave ripples, the words can be reassembled into novel sentences to match the extended sequences of discrete segments of travel paths, situations, or events (Chapter 8). This "chunking" or parsing role of chained sharp wave ripples

31. Like head-direction cells, border cells are also in place before spontaneous ambulation. Grid cells have irregular and variable fields until the fourth week, after the emergence of place cells. Together, they form a functional map at the time when rat pups leave the nest for the first time in their life, and the maps get fine-tuned thereafter (Langston et al., 2010; Wills et al., 2014; Bjerknes et al., 2014). Muessig et al. (2016) state that: "What drives remapping in young pups remains an open question." Perhaps nothing really "drives" remapping, it is a matching process between already existing maps and ambulatory exploration.

32. For example, cat = xx.u. By building an initial set this way, when next time we have a word in mind (dog), we should not only find another random word (n-zk) but somehow indicate a relationship between them. This analogy has a major weakness as well. The initially meaningless neuronal words in the brain may already have preexisting relationships to each other. You can call it prejudice. These preconfigured constraints bias how the pages open up in our hypothetical dictionary.

allows the flexible generation of a large number of combinations from a finite number of preformed neuronal words.[33]

Overall, learning with neurons involves a synthesis of preexisting neuronal events, which only modestly and transiently affects network dynamics. In each learning episode, a small number of the many possible preexisting neuronal trajectories acquire behavioral meaning and thus start to represent a particular constellation of relationships. I have used spatial learning as an example because the most quantitative experimental data are available in this field. The same process likely applies when neuronal trajectories are matched to events and situations. This view echoes the idea that perception is hypothesis-testing in the brain.[34] Matching is the brain's attempt to pull out the best-fitting neuronal trajectory in any situation from its preformed vast reservoir and refine it only when mismatch occurs.

SUMMARY

The Weber-Fechner law describes not only weight, loudness, brightness, and odorant intensity, but also more abstract dimensions, such as representation of sizes, distances, owned territory, intervals, durations, numbers, and other symbolic representation of quantities. In this chapter and the previous one, I proposed that these mental metrics are based on a skewedly organized brain structure and dynamics. Behavior-based calibration of perceptions and even abstract representations stem directly from the constraints of a preconfigured brain. This view is different from the philosophical idea of isomorphism,[35] which assumes that things in the brain correspond to shapes and properties in the world. Instead, the nervous system may have evolved to mimic the statistical probabilities of the physical world and thus become an efficient predictor

33. For more details on concatenation of neuronal "chunks" by sharp wave ripples, see Buzsáki (2015) and Pfeiffer (2018). Sharp wave ripples during post-experience sleep are also critical for reinstating the default network dynamics after experience-induced perturbation (Grosmark et al., 2012). After learning and even after plasticity-inducing electrical stimulation, such as long-term potentiation, synapses may get stronger, and some neurons increase their firing rates but only at the expense of others. Within the same brain state, the total synaptic weights and the mean excitability of the system remains constant over time (Dragoi et al., 2003).

34. Helmholtz (1866/1962); Gregory (1980). Bayesian models of the brain express similar sentiments Friston (2010, 2012). However, the Bayesian brain falls short in explaining how the brain creates new knowledge (Friston and Buzsáki, 2016).

35. Ullman (1980); Biederman (1987).

of events. As a result, neurophysiological and perceptual brain dynamics, both spanning several orders of magnitude, share a common mathematical foundation: the log rule.

Skewed distributions at multiple levels of brain organization come with both advantages and disadvantages. The tails of these quantitative distributions have apparently distinct qualitative features, which we describe by discrete words, such as familiar and novel, rigid and plastic, good-enough and precise.[36] Yet every novel situation contains elements of familiarity. Similarly, the plastic versus rigid division is really a continuum. Brain correlates of newly acquired experience are not created in the sense of inventing new words from nothing to an ever-expanding vocabulary. Instead, the preformed brain is a dictionary in which initially there is no meaning assigned to its preexisting words and sentences supported by the numerous neuronal trajectories that brain networks perpetually generate. The behavioral significance or meaning of neuronal words in the brain dictionary is acquired through exploration. Thus, experience is primarily a process of matching preexisting neuronal dynamics to events in the world.

36. The Babel of terms describing the physiological attributes of neurons in the hippocampal-entorhinal system (such as place cells, time cells, grid cells, head direction cells, boundary vector cells, border cells, object cells, object vector cells, distance cells, reward cells, concept cells, view cells, speed cells, memory cells, goal cells, goal direction cells, splitter cells, prospective cells, retrospective cells, island cells, ocean cells, band cells, object cells, pitch cells, trial order cells, etc.) might be explained by the apparent distinctiveness of a few sets of continuous distributions. Relations are often quantitative rather than qualitative. Naming elements of a distribution creates the illusion that they are distinct (i.e., qualitatively different).